with constant variance). We use notation (A, B) for the concatenation of two matrices A, B. We use the notation $a_t, b_t \perp x_t, y_t$ to signify that random variables a_t and b_t are uncorrelated with random variables x_t and y_t.

II. GIVs: Main Results

In this section, we formally introduce GIVs at a fairly high level of generality. Section III.A contains an instructive simple case. We also establish consistency of the estimators and derive their asymptotic distributions.

A. *The Baseline GIV*

Setup.—Our goal is to estimate the scalar parameters ψ and ϕ^d in the following prototypical model:

$$p_t = \psi y_{St} + C_t^p m^p + \varepsilon_t, \tag{1}$$

$$y_{it} = \phi^d p_t + C_{it}^y m^y + \lambda_i \eta_t + u_{it}, \tag{2}$$

where $y_{St} := \Sigma_i S_i Y_{it}$, with S_i representing the weight on entity $i \in \{1, \dots, N\}$ and with $\Sigma_i S_i = 1$. An economic interpretation is that of demand and supply for a given good—for example, oil.[2] Each entity's log demand y_{it} depends on the log price p_t of that good as in (2), with demand elasticity ϕ^d, and some shocks. The price in turn depends on aggregate demand y_{St}, which is the size-weighted demand, with a sensitivity ψ that is the inverse of the supply elasticity. We impose that $\psi\phi^d \neq 1$, which ensures the existence of a unique solution of our system (1)–(2).

The econometrician observes S_i, p_t, y_{it}, and the controls C_t^p and C_{it}^y, which include constants and entity fixed effects; C_t^y is the vector of C_{it}^y's. The parameters ψ, ϕ^d, m^p, and m^y have to be estimated, and we are chiefly interested in ψ and ϕ^d. The shocks ε_t, η_t, and $u_t = (u_{it})_{i=1,\dots,N}$ are not observed. The demand-side aggregate shocks are captured by a factor model of dimension r, $\lambda_i \eta_t = \Sigma_{f=1}^r \lambda_i^f \eta_t^f$, so that λ_i and η_t have dimensions $1 \times r$ and $r \times 1$, respectively. We assume that the first component of λ_i is one, and the other components have been normalized to have mean zero, so we can write $\lambda_i = (1, \check{\lambda}_i)$—this is simply a normalization. We call $\lambda = (\lambda_i)_{i=1,\dots,N}$ and $\check{\lambda} = (\check{\lambda}_i)_{i=1,\dots,N}$ the $N \times r$ and $N \times (r-1)$ matrices collecting the λ_i's and $\check{\lambda}_i$'s, respectively. The normalization implies that $\iota'\check{\lambda} = 0_{1\times(r-1)}$.

Throughout the paper, the number of entities N is fixed, and we study estimators in the limit where the number of periods $T \to \infty$. We view the data $(y_t, p_t, C_t^y, C_t^p)_{t=1,\dots,T}$ as sampled independent and identically distributed

[2] Appendix section D.2 contains a simple economic model leading to this structure.

(i.i.d.) across periods,[3] and we take the vector of weights S_i to be fixed (i.e., nonrandom).

Assumptions made throughout the paper.—We maintain the following assumptions throughout the paper. All random variables have finite fourth moments. The aggregate and idiosyncratic shocks, η_t, ε_t, and u_t, are uncorrelated across periods and have mean zero. The controls C_t^p and C_t^y are mutually uncorrelated with η_t and ε_t: $C_t^p, C_t^y \perp \eta_t, \varepsilon_t$.

The central assumption is that shocks u_{it} are idiosyncratic—that is, they are uncorrelated with the aggregate shocks, η_t and ε_t, and with controls, C_t^y and C_t^p. Formally,

$$u_t \perp \eta_t, \varepsilon_t, C_t^y, C_t^p. \tag{3}$$

Assumptions made for expositional convenience.—We sometimes make assumptions that simplify the exposition or the reasoning, typically with minimal impact on the economics. We indicate in the remainder of the paper when we make these auxiliary assumptions.

ASSUMPTION 1. The λ_i's are known.

Assumption 1 is weaker than it seems. For instance, it holds when the λ_i's simply have a parametric form, $\lambda_i = X_i\dot{\lambda}$, where λ_i and X_i are r-dimensional row vectors and $\dot{\lambda}$ is an $r \times r$ matrix. The vector of characteristics X_i is observed. This implies that $\lambda = X\dot{\lambda}$, where λ and X are $N \times r$ matrices. This assumption simplifies the exposition and reasoning, but it will be relaxed in section IV.B. As one can change η_t into $\dot{\lambda}\eta_t$, one might normalize $\dot{\lambda} = I_r$.

The following three assumptions also simplify the exposition, and we relax each of them later in the paper.

ASSUMPTION 2. The controls have the shape $C_{it}^y = \lambda_i c_t^y$.

ASSUMPTION 3. The u_{it}'s are uncorrelated across i's and homoskedastic with variance $\sigma_u^2 > 0$.

ASSUMPTION 4. The u_{it}'s are heteroskedastic with variance-covariance matrix $V^u = \mathbb{E}[u_t u_t']$ that is nonsingular and constant over time.

The core idea of GIVs.—The idea of GIVs is to use idiosyncratic shocks u_{it} as instruments. We construct GIVs as follows, where we impose assumption 2 for now. Suppose that we have a set of weights $\Gamma \in \mathbb{R}^N$ orthogonal to factor loadings λ but not to the size vector S:

$$\Gamma'\lambda = 0, \qquad \Gamma' S \neq 0. \tag{4}$$

This is possible when S is not spanned by the loadings λ.[4] We provide a concrete and optimal construction of Γ in (17). Then the GIV is defined as

[3] One could extend this—e.g., to a setting with stochastic volatility—but we keep the i.i.d. assumption for simplicity.

[4] That is, when there is no vector b such that $S = \lambda b$.

$$z_t := \Gamma' y_t = \sum_{i=1}^{N} \Gamma_i y_{it}. \tag{5}$$

Hence, the GIV z_t is constructed from observables, y_{it}. As $\Gamma'\lambda = 0$, we have $\Gamma'\iota = 0$ (recall that $\iota = (1, \ldots, 1)'$ and that the first column vector of λ is a one). The key observation, using assumption 2 for now,[5] is that $z_t := \Gamma' y_t = \Gamma'(\iota\phi^d p_t + \lambda c_t^y + \lambda\eta_t + u_t) = \Gamma' u_t$, so

$$z_t = \Gamma' u_t. \tag{6}$$

As a result, the GIV z_t is a linear combination of idiosyncratic shocks. The GIV satisfies the exogeneity condition by (3):

$$\text{exogeneity:} \quad z_t \perp \eta_t, \varepsilon_t. \tag{7}$$

The GIV also satisfies the relevance condition because $\Gamma' S \neq 0$:

$$\text{relevance:} \quad \mathbb{E}[y_{St} z_t] \neq 0. \tag{8}$$

Then, (1) implies the moment condition:[6]

$$\mathbb{E}[(p_t - \psi y_{St}) z_t] = 0, \tag{9}$$

and we can estimate ψ using the GIV z_t as an instrument, as $\psi = \mathbb{E}[p_t z_t]/\mathbb{E}[y_{St} z_t]$. Its natural empirical counterpart is the estimator $\psi_T^e = \Sigma_{t=1}^T p_t z_t / \Sigma_{t=1}^T y_{st} z_t$.

To estimate ϕ^d, we take a vector $\tilde{E}$ such that

$$\mathbb{E}[u_{\tilde{E}t} u_{\Gamma t}] = 0, \qquad \iota' \tilde{E} = 1. \tag{10}$$

For instance, with homoskedastic residuals, we can take $\tilde{E}_i = 1/N$, which leads us to call $\tilde{E}$ a "quasi-equal weight" vector more generally.[7] As $y_{\tilde{E}t} := \Sigma_i \widetilde{E}_i y_{it} = \phi^d p_t + C_{\tilde{E}t}^y m^y + \lambda_{\tilde{E}} \eta_t + u_{\tilde{E}t}$, with $u_{\tilde{E}t} := \Sigma_i \widetilde{E}_i u_{it}$, (10) implies that

$$\mathbb{E}[(y_{\tilde{E}t} - \phi^d p_t) z_t] = 0. \tag{11}$$

The relevance condition is then

$$\text{relevance:} \quad \mathbb{E}[p_t z_t] \neq 0. \tag{12}$$

This condition is satisfied if $\psi \neq 0$—that is, when demand shocks influence the price. Then we can estimate ϕ^d using z_t as an instrument, via $\phi^d = \mathbb{E}[y_{\tilde{E}t} z_t]/\mathbb{E}[p_t z_t]$. Its empirical counterpart is $\phi_T^{d,e} = \Sigma_{t=1}^T y_{\tilde{E}t} z_t / \Sigma_{t=1}^T p_t z_t$. The following proposition establishes the consistency of the GIV estimator.

[5] Alternatively, the reader is encouraged to think of the even simpler case where there are no controls, $C_t^y = 0$.

[6] The error term, $p_t - \psi y_{St}$, includes aggregate shocks and controls. This moment condition holds, even though it is more efficient to account for controls in estimating ψ; see sec. II.C.

[7] See (57) for a proof. The heteroskedastic case will lead to (54).

PROPOSITION 1 (Consistency of the GIV estimator). Let assumptions 1 and 2 hold, and assume that $\mathbb{E}[u_{St}u_{\Gamma t}] \neq 0$. Define the GIV estimators $\psi_T^e = \Sigma_{t=1}^T p_t z_t / \Sigma_{t=1}^T y_{St} z_t$ and $\phi_T^{d,e} = \Sigma_{t=1}^T y_{\bar{E}t} z_t / \Sigma_{t=1}^T p_t z_t$. Then, ψ_T^e is a consistent estimator of ψ: as $T \to \infty$, $\psi_T^e \xrightarrow{\text{a.s.}} \psi$. Likewise, if $\psi \neq 0$, then $\phi_T^{d,e}$ is a consistent estimator of ϕ^d: as $T \to \infty$, $\phi_T^{d,e} \xrightarrow{\text{a.s.}} \phi^d$.

To think about the variance of the estimators and—not coincidentally—the economics, we define

$$M := \frac{1}{1 - \psi\phi^d}, \qquad \mu := \psi M. \tag{13}$$

Those quantities are the pass-through from idiosyncratic and aggregate shocks to the aggregate quantity and the price, respectively; see (33) and (34).

PROPOSITION 2 (Variance of the GIV estimator). Let assumptions 1 and 2 hold, and assume that $\mathbb{E}[u_{St}u_{\Gamma t}] \neq 0$. Assume that $\varepsilon_t^\psi := C_t^p m^p + \varepsilon_t$ is homoskedastic conditionally on u_t with variance $\sigma^2_{\varepsilon^\psi}$. Then, as $T \to \infty$,

$$\sqrt{T}(\psi_T^e - \psi) \xrightarrow{d} N(0, \sigma_\psi^2), \qquad \sigma_\psi = \frac{\mathbb{E}\left[u_{\Gamma t}^2\right]^{1/2}}{|\mathbb{E}[u_{St}u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\psi}}{|M|}. \tag{14}$$

Likewise, if $\mu \neq 0$, then, assuming that $\varepsilon_t^\phi := C_{\bar{E}t}^y m^y + \lambda_{\bar{E}}\eta_t + u_{\bar{E}t}$ has variance $\sigma^2_{\varepsilon^\phi}$ conditionally on u_t,

$$\sqrt{T}(\phi_T^{d,e} - \phi^d) \xrightarrow{d} N\left(0, \sigma^2_{\phi^d}\right), \qquad \sigma_{\phi^d} = \frac{\mathbb{E}\left[u_{\Gamma t}^2\right]^{1/2}}{|\mathbb{E}[u_{St}u_{\Gamma t}]|} \frac{\sigma_{\varepsilon^\phi}}{|\mu|}. \tag{15}$$

We provide further economic intuition for the asymptotic variances of the GIV estimators in section III.A. The condition $\mathbb{E}[u_{St}u_{\Gamma t}] \neq 0$ is generically true. For instance, with homoskedastic u_{it}, $\mathbb{E}[u_{St}u_{\Gamma t}] = \sigma_u^2 \Gamma' S$; it is guaranteed by (4).

B. *Optimal GIV Weights and the Precision of the GIV Estimator*

We now derive the optimal GIV weights Γ that minimize the asymptotic variances σ_ψ^2 and $\sigma^2_{\phi^d}$ in proposition 2. For this, we use the $N \times N$ projection matrix Q that is orthogonal to the factor loadings—that is, $Q\lambda = 0$:

$$Q := I - \lambda(\lambda'\lambda)^{-1}\lambda'. \tag{16}$$

PROPOSITION 3 (Optimal weights Γ for the GIV $y_{\Gamma t}$). Let assumptions 1, 2, and 3 hold. The optimal weights

$$\Gamma^{*\prime} = S'Q \tag{17}$$

minimize the asymptotic variances σ_ψ^2 and $\sigma^2_{\phi^d}$ of the GIV estimators in proposition 2, with $\mathbb{E}[u_{\Gamma t}^2]/\mathbb{E}[u_{St}u_{\Gamma t}]^2 = 1/\mathbb{E}[u_{\Gamma^* t}^2]$. For any other Γ that is